F18CE2 - MV Calculus and Real Analysis B: Solutions 8

1 Let $S_n = \sum_{r=1}^{n} a_r$. Then $\sum_{r=1}^{\infty} a_r$ converges if the sequence (S_n) converges.

If $a_n = x_n - x_{n+1}$, then

$$S_n = \sum_{r=1}^{n}(x_r - x_{r+1}) = x_1 - x_2 + x_2 - x_3 + x_3 - x_4 + \ldots x_n - x_{n+1} = x_1 - x_{n+1}.$$

Since (x_n) is a convergent sequence we can define $L = \lim_{n\to\infty} x_n$, and we then have

$$\lim_{n\to\infty} S_n = x_1 - \lim_{n\to\infty} x_n = x_1 - L,$$

so the series $\sum_{r=1}^{\infty} a_r$ converges with sum $x_1 - L$.

2

$$S_n = \sum_{r=1}^{n} \ln(1 + 1/r) = \sum_{r=1}^{n}(\ln(r+1) - \ln r) = \ln(n+1) \to \infty \quad \text{as } n \to \infty.$$

Since the sequence (S_n) diverges, the series diverges.

3 (a) For $a_n = (-1)^n$, the limit $\lim_{n\to\infty} a_n$ does not exist so the series diverges.

(b) For $a_n = \dfrac{2n+1}{n+3}$, $\lim_{n\to\infty} a_n = 2$ so the series diverges.

4 Comparison Test: Suppose that $0 \leqslant a_r \leqslant b_r$ for all r. Then:

$\sum_{r=1}^{\infty} b_r$ converges $\implies$ $\sum_{r=1}^{\infty} a_r$ converges;

$\sum_{r=1}^{\infty} a_r$ diverges $\implies$ $\sum_{r=1}^{\infty} b_r$ diverges.

Since $a_r \geqslant 0$, $\dfrac{a_r}{1+a_r} \leqslant a_r$.

Since $\sum_{r=1}^{\infty} a_r$ converges, the series $\sum_{r=1}^{\infty} \dfrac{a_r}{1+a_r}$ also converges, by the comparison test.

5 (a) For all $r \geqslant 1$,

$$\frac{r^2}{r^3+1} \geqslant \frac{r^2}{r^3+r^3} = \frac{1}{2r}.$$

Since $\sum_{r=1}^{\infty} \dfrac{1}{r}$ diverges, the series $\sum_{r=1}^{\infty} \dfrac{r^2}{r^3+1}$ diverges by the comparison test.

(b) For all $r \geqslant 1$,

$$\frac{r}{r^3+1} \leqslant \frac{r}{r^3} = \frac{1}{r^2}.$$

Since $\sum_{r=1}^{\infty} \dfrac{1}{r^2}$ converges, the series $\sum_{r=1}^{\infty} \dfrac{r}{r^3+1}$ converges by the comparison test.

(c) $a_r \geqslant 0$ for all r and

$$\frac{(r+1)\sin^2 r}{r^3+3r+1} \leqslant \frac{r+r}{r^3} = \frac{2}{r^2}.$$

Since $\displaystyle\sum_{r=1}^{\infty} \frac{1}{r^2}$ converges, the series $\displaystyle\sum_{r=1}^{\infty} \frac{(r+1)\sin^2 r}{r^3+3r+1}$ converges.

6 $(a_r - b_r)^2 \geqslant 0$, so for all r,

$$a_r^2 + b_r^2 \geqslant 2a_r\, b_r\,.$$

Since $\displaystyle\sum_{r=1}^{\infty}(a_r^2+b_r^2)$ converges, by the comparison test the series $\displaystyle\sum_{r=1}^{\infty} a_r\, b_r$ also converges.

Example: the series $\displaystyle\sum_{r=1}^{\infty} \frac{1}{r}$ diverges, but the series $\displaystyle\sum_{r=1}^{\infty} \frac{1}{r^2}$ converges.

7 (a) For all r, $Ar^{-1} \leqslant a_r$. Since $\displaystyle\sum_{r=1}^{\infty} \frac{1}{r}$ diverges, by the comparison test $\displaystyle\sum_{r=1}^{\infty} a_r$ diverges.

(b) For all r, $a_r^2 \leqslant B^2 r^{-2}$. Since $\displaystyle\sum_{r=1}^{\infty} \frac{1}{r^2}$ converges, by the comparison test $\displaystyle\sum_{r=1}^{\infty} a_r^2$ converges.

(c) For all r, $r^{-1}\sqrt{a_r} \leqslant \sqrt{B} r^{-3/2}$. Since $\displaystyle\sum_{r=1}^{\infty} \frac{1}{r^{3/2}}$ converges, $\displaystyle\sum_{r=1}^{\infty} r^{-1}\sqrt{a_r}$ converges.

8 Ratio Test: Suppose that for all $r \geqslant 1$, $a_r > 0$ and $\displaystyle\lim_{r\to\infty} \frac{a_{r+1}}{a_r} = L$.

Then the series $\displaystyle\sum_{r=1}^{\infty} a_r$ converges if $L < 1$ and diverges if $L > 1$.

(a) $\displaystyle\frac{a_{r+1}}{a_r} = \frac{6^{r+1}\, r!}{(r+1)!\, 6^r} = \frac{6}{r+1} \to 0$ as $r \to \infty$ so the series converges.

(b) $\displaystyle\frac{a_{r+1}}{a_r} = \frac{(r+1)\, 3^{r+1}\, 4^r}{r\, 4^{r+1}\, 3^r} = \frac{3(r+1)}{4r} \to \frac{3}{4}$ as $r \to \infty$ so the series converges.

(c) $\displaystyle\frac{a_{r+1}}{a_r} = \frac{5^{r+1}\, 4^r\, r}{(r+1)\, 4^{r+1}\, 5^r} = \frac{5r)}{4(r+1)} \to \frac{5}{4}$ as $r \to \infty$ so the series diverges.

9 (a)

$$|a_r| = \left|\frac{\cos r}{r^2+1}\right| \leqslant \left|\frac{1}{r^2+1}\right| \leqslant \frac{1}{r^2}.$$

Since $\displaystyle\sum_{r=1}^{\infty} 1/r^2$ converges, by the comparison test, $\displaystyle\sum_{r=1}^{\infty} |a_r|$ converges.

By the absolute convergence test, the series $\displaystyle\sum_{r=1}^{\infty} \frac{\cos r}{r^2+1}$ converges.

(b)

$$|a_r| = \left|\frac{r^2 \sin r}{r^4+3r^2+1}\right| \leqslant \left|\frac{r^2}{r^4+3r^2+1}\right| \leqslant \frac{r^2}{r^4} = \frac{1}{r^2}.$$

Since $\displaystyle\sum_{r=1}^{\infty} 1/r^2$ converges, by the comparison test, $\displaystyle\sum_{r=1}^{\infty} |a_r|$ converges.

By the absolute convergence test, the series $\displaystyle\sum_{r=1}^{\infty} \frac{r^2 \sin r}{r^4+3r^2+1}$ converges.

(c) $$\frac{|a_{r+1}|}{|a_r|} = \frac{(r+1)^2\,2^r}{r^2\,2^{r+1}} = \frac{(r+1)^2}{2r^2} \to \frac{1}{2} \quad \text{as } r \to \infty.$$

By the ratio test, $\displaystyle\sum_{r=1}^{\infty} |a_r|$ converges, so by the absolute convergence test $\displaystyle\sum_{r=1}^{\infty} \frac{r^2\,(-1)^r}{2^r}$ converges.

10 (a) Let $b_r = \dfrac{(-1)^r r^3 x^r}{3^r}$. Then,

$$\left|\frac{b_{r+1}}{b_r}\right| = \frac{(r+1)^3\,|x|}{3r^3} \quad \text{so} \quad \lim_{r\to\infty}\left|\frac{b_{r+1}}{b_r}\right| = |x|/3.$$

So by the ratio test $\displaystyle\sum_{r=1}^{\infty} |b_r|$ converges for $|x| < 3$ and diverges for $|x| > 3$.

So the radius of convergence is $R = 3$.

(b) Let $b_r = \dfrac{x^r}{6^r\sqrt{r^2+2r}}$. Then,

$$\frac{b_{r+1}}{b_r} = \frac{x^{r+1}\,6^r\,\sqrt{r^2+2r}}{6^{r+1}\,x^r\,\sqrt{(r+1)^2+2r+2}} = \frac{x\sqrt{r^2+2r}}{6\sqrt{(r+1)^2+2r+2}} \quad \text{so} \quad \lim_{r\to\infty}\left|\frac{b_{r+1}}{b_r}\right| = \frac{|x|}{6}.$$

So by the ratio test $\displaystyle\sum_{r=1}^{\infty} |b_r|$ converges for $|x| < 6$ and diverges for $|x| > 6$, and so the radius of convergence is $R = 6$.

When $x = R = 6$ the series is $\displaystyle\sum_{r=1}^{\infty} \frac{1}{\sqrt{r^2+2r}}$. For $r \geqslant 1$,

$$\frac{1}{\sqrt{r^2+2r}} \geqslant \frac{1}{\sqrt{r^2+2r^2}} = \frac{1}{\sqrt{3}\,r}$$

and $\displaystyle\sum_{r=1}^{\infty} \frac{1}{r}$ diverges. Hence the given series diverges when $x = R = 6$ by the comparison test.

When $x = -R = -6$ the series is $\displaystyle\sum_{r=1}^{\infty} (-1)^r \frac{1}{\sqrt{r^2+2r}}$ which converges by the alternating series test.